

Unit 6.6: Cubic & Cube Root Functions
Mixed Transformations Practice — Advanced

Name: KEY
Period: _____

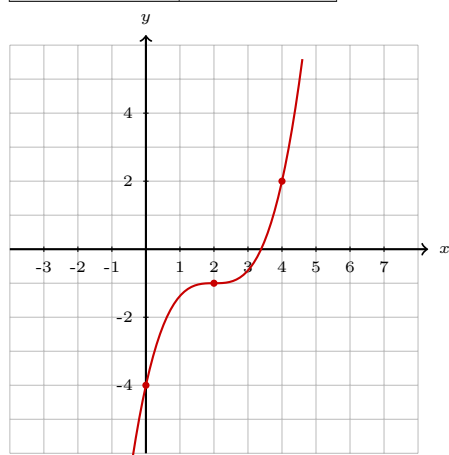
Reminder: $f(x) = a(b(x-h))^3 + k$ or $f(x) = a\sqrt[3]{b(x-h)} + k$ where b controls horizontal stretch/compression

1. $f(x) = 3\left(\frac{1}{2}(x-2)\right)^3 - 1$

$a = 3$ $b = 0.5$ $h = 2$ $k = -1$

Center: $(2, -1)$

Parent x^3	New
$(-1, -1)$	$(0, -4)$
$(0, 0)$	$(2, -1)$
$(1, 1)$	$(4, 2)$
$(2, 8)$	$(6, 23)$
$(-2, -8)$	$(-2, -25)$



x -int: ≈ 3.38

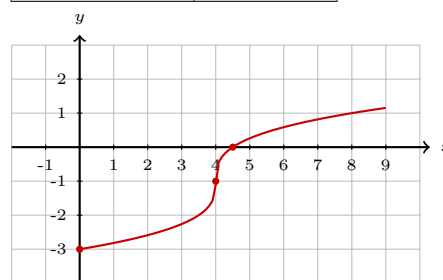
y -int: $(0, -4)$

2. $f(x) = \sqrt[3]{2(x-4)} - 1$

$a = 1$ $b = 2$ $h = 4$ $k = -1$

Center: $(4, -1)$

Parent $\sqrt[3]{x}$	New
$(-8, -2)$	$(0, -3)$
$(-1, -1)$	$(3.5, -2)$
$(0, 0)$	$(4, -1)$
$(1, 1)$	$(4.5, 0)$
$(8, 2)$	$(8, 1)$



x -int: $(4.5, 0)$

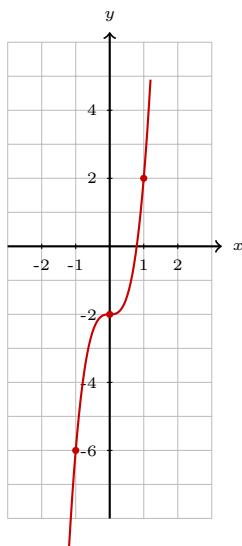
y -int: $(0, -3)$

3. $f(x) = \frac{1}{2}(2x)^3 - 2 = 4x^3 - 2$

$a = 0.5$ $b = 2$ $h = 0$ $k = -2$

Center: $(0, -2)$

Parent x^3	New
$(-2, -8)$	$(-1, -6)$
$(-1, -1)$	$(-0.5, -2.5)$
$(0, 0)$	$(0, -2)$
$(1, 1)$	$(0.5, -1.5)$
$(2, 8)$	$(1, 2)$



x -int: ≈ 0.79

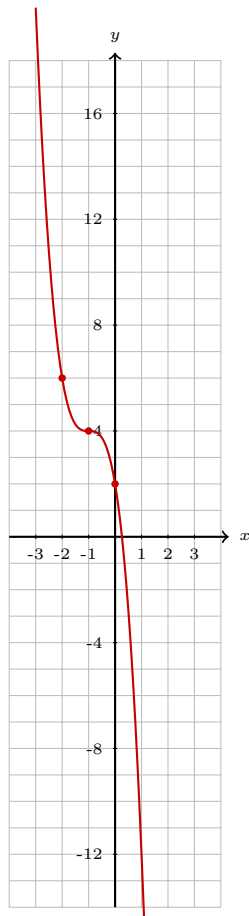
y -int: $(0, -2)$

5. $f(x) = -2(x + 1)^3 + 4$

$a = -2$ $b = 1$ $h = -1$ $k = 4$

Center: $(-1, 4)$

Parent x^3	New
$(-2, -8)$	$(-3, 20)$
$(-1, -1)$	$(-2, 6)$
$(0, 0)$	$(-1, 4)$
$(1, 1)$	$(0, 2)$
$(2, 8)$	$(1, -12)$



x -int: ≈ 0.26

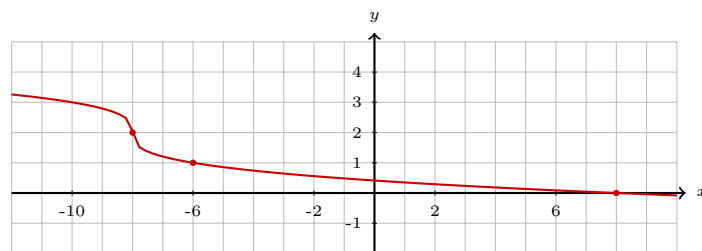
y -int: $(0, 2)$

4. $f(x) = -\sqrt[3]{\frac{1}{2}(x + 8)} + 2$

$a = -1$ $b = 0.5$ $h = -8$ $k = 2$

Center: $(-8, 2)$

Parent $\sqrt[3]{x}$	New
$(-8, -2)$	$(-24, 4)$
$(-1, -1)$	$(-10, 3)$
$(0, 0)$	$(-8, 2)$
$(1, 1)$	$(-6, 1)$
$(8, 2)$	$(8, 0)$



x -int: $(8, 0)$

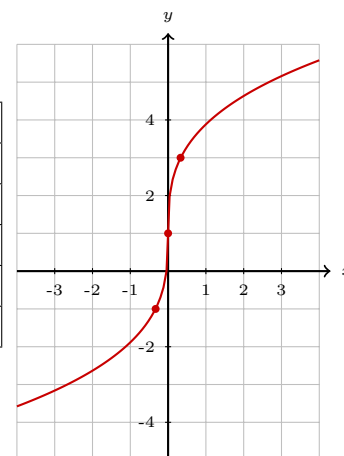
y -int: $\approx (0, 0.41)$

6. $f(x) = 2\sqrt[3]{3x} + 1$

$a = 2$ $b = 3$ $h = 0$ $k = 1$

Center: $(0, 1)$

Parent $\sqrt[3]{x}$	New
$(-8, -2)$	$(-2.6, -3)$
$(-1, -1)$	$(-0.3, -1)$
$(0, 0)$	$(0, 1)$
$(1, 1)$	$(0.3, 3)$
$(8, 2)$	$(2.6, 5)$

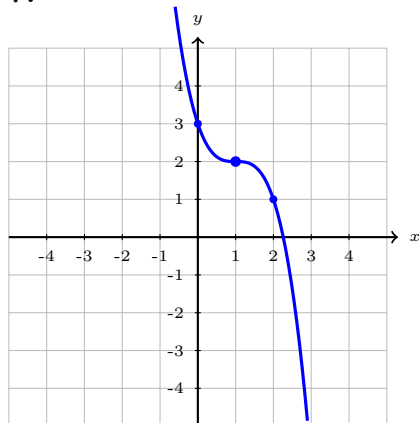


x -int: ≈ -0.04

y -int: $(0, 1)$

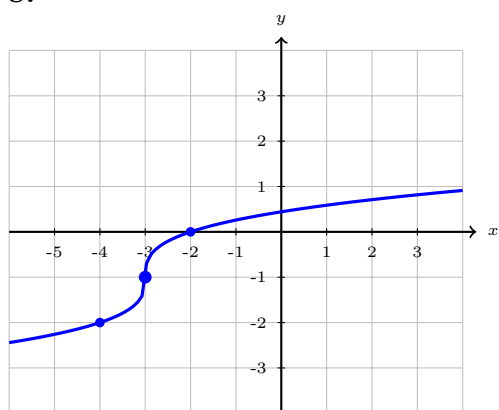
Write the equation from each graph:

7.



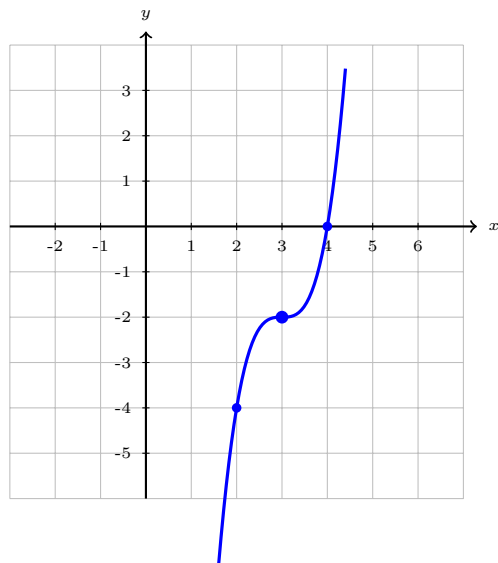
$$f(x) = -(x - 1)^3 + 2$$

8.



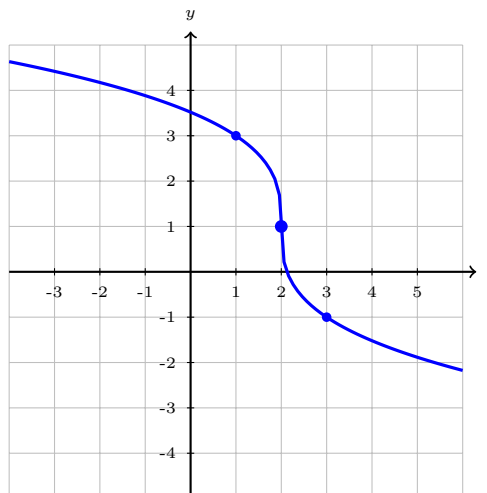
$$f(x) = \sqrt[3]{x + 3} - 1$$

9.



$$f(x) = 2(x - 3)^3 - 2$$

10.



$$f(x) = -2\sqrt[3]{x - 2} + 1$$

Write the equation from the description:

11. A cubic function that is vertically stretched by a factor of 2, horizontally compressed by a factor of $\frac{1}{3}$, reflected over the x -axis, shifted left 4, and shifted up 5.

$$f(x) = -2(3(x + 4))^3 + 5$$

12. A cube root function that is vertically compressed by a factor of $\frac{1}{2}$, horizontally stretched by a factor of 2, shifted right 6, and shifted down 3.

$$f(x) = \frac{1}{2}\sqrt[3]{\frac{1}{2}(x - 6)} - 3$$

Error Analysis:

13. Marcus says the function $f(x) = \sqrt[3]{2(x - 3)} + 1$ has a horizontal stretch by a factor of 2. Explain his error and give the correct transformation.

Marcus confused b with the scale factor. Since $b = 2$ and $|b| > 1$, it is a horizontal **compression** by a factor of $\frac{1}{2}$.

14. Destiny graphed $f(x) = (x + 4)^3 - 2$ with a center point at $(4, -2)$. What mistake did she make?

Destiny forgot to flip the sign of h . The term $(x + 4)$ means $h = -4$. The center should be at $(-4, -2)$.

Compare and Contrast:

15. Without graphing, explain how $f(x) = 2x^3$ and $g(x) = (2x)^3$ are different. Which one is steeper? Justify your answer.

$g(x) = (2x)^3 = 8x^3$. Since $8 > 2$, $g(x)$ has a larger vertical stretch factor, making it much steeper/narrower than $f(x)$.

16. The functions $f(x) = 8x^3$ and $g(x) = (2x)^3$ produce identical graphs. Explain why this is true algebraically.

Using exponent rules, $(2x)^3 = 2^3 \cdot x^3 = 8x^3$. Therefore, a horizontal compression by $1/2$ (inside the cube) is equivalent to a vertical stretch by 8.

Challenge Problems — For students who finish early

17. A cubic function has its center point at $(1, 3)$ and passes through the point $(2, 5)$. Write the equation of the function.

$$\begin{aligned}y &= a(x - 1)^3 + 3 \implies 5 = a(2 - 1)^3 + 3 \implies 5 = a(1) + 3 \implies a = 2. \\f(x) &= 2(x - 1)^3 + 3\end{aligned}$$

18. A cube root function has its center point at $(-2, 4)$. If $f(6) = 8$, find the value of a and write the equation.

$$\begin{aligned}y &= a\sqrt[3]{x + 2} + 4 \implies 8 = a\sqrt[3]{6 + 2} + 4 \implies 4 = a\sqrt[3]{8} \implies 4 = 2a \implies a = 2. \\f(x) &= 2\sqrt[3]{x + 2} + 4\end{aligned}$$

19. A cubic function satisfies $f(0) = 1$ and $f(2) = -15$ with a center point at $(1, 3)$.

(a) Use one of the points to find the value of a .

$$\text{Using } (0, 1) \text{ and center } (1, 3): 1 = a(0 - 1)^3 + 3 \implies 1 = -a + 3 \implies a = 2.$$

(b) Verify your answer using the other point. Show your work.

$$\text{Check } (2, -15): -15 = 2(2 - 1)^3 + 3 \implies -15 = 2(1) + 3 \implies -15 \neq 5.$$

Note to Teacher: The problem text contains a typo. For symmetry with center $(1, 3)$ and point $(0, 1)$, $f(2)$ should be 5, not -15 .

(c) Write the final equation: $f(x) = 2(x - 1)^3 + 3$